

New boron compounds as additives for lithium polymer electrolytes

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Abstract

A new class of difluoroalkoxyborane compounds ($[R_n\text{OBF}_2]_2$) containing oligooxyethylene groups of various molecular weight in the form of a methyl monoether ($R_n = \text{CH}_3(\text{OCH}_2\text{CH}_2)_n$, $n = 1, 2, 3$ and 7) has been obtained in the reaction of BF_3 etherate with appropriate glycols. ^1H , ^{11}B and ^{19}F NMR spectral analysis of the derivatives obtained was carried out and the properties as Lewis acids of these derivatives have been compared with that of corresponding trialkoxyboranes and boron trifluoride in reaction with pyridine. The strength of the interaction of $[R_2\text{OBF}_2]_2$ with the differing in “hardness” anions of various lithium salts has been analyzed on the basis of NMR spectra. The $[R_n\text{OBF}_2]_2$ obtained were used as additives for polymer electrolytes containing PEO as polymer matrix and various lithium salts at an equimolar ratio of the boron compound to salt. The highest ionic conductivities, in the order 10^{-5} to $10^{-4} \text{ S cm}^{-1}$ at 20–70 °C, were achieved for systems containing LiI and $\text{LiN}(\text{CF}_3\text{SO}_2)_2$. The lithium transference number (t_+) values, determined by the electrochemical method by steady-state technique for LiF and LiCF_3SO_3 are in the 0.6–0.8 range.

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1. Introduction

Intense studies are being carried out on the utilization of solvent-free polymeric materials as separators in rechargeable lithium batteries [1,2]. Problems connected with the evaporation of low molecular weight solvent and its reactivity towards the anode are thus avoided. However, despite many studies, no SPE has yet been obtained of which the ionic conductivity would be sufficient for practical application. From studies carried out until now it appears that poly(ethylene oxide) (PEO) is characterized by the best properties as a solid solvent for lithium salts. Moreover, a majority of works are concerned with the modification of electrolytes based on PEO, in order to increase the ionic conductivity while maintaining suitable mechanical properties and achieving such operating conditions in which completely or partly the anionic current is eliminated. The motion of anions during operation of the battery causes the formation of polarization layers worsening the battery parameters. Depending on the measurement method and the salt concentration, the lithium

transference number values of SPE based on PEO lie in the 0–0.5 range [3]. One of the conceptions of achieving this goal consists in the introduction to the electrolyte of an additive, which would increase the ambient temperature ionic conductivity and the lithium ions transference number by interaction with the salt anion. Such an interaction results in the facilitation of the salt dissociation, decrease in the concentration of ion pairs and decrease in the mobility of anions. Examples are reported in the literature of the application of various additives of Lewis acid properties, such as nanometric inorganic fillers, which on the grain surface bear acidic centers interacting with the basic salt anions, which leads to favorable changes in the properties of composite electrolytes, such as increase in conductivity, chemical stability in contact with the lithium electrode and lithium transference number [4–8]. The other method consists in the introduction of so-called anion receptors, such as calixarenes [9,10] or calixpyrroles [11], causing a decrease in the share of ion pairs and immobilization of anions, resulting in an increase in the cation transference number. Aluminum [12–14] or boron derivatives of Lewis acid properties capable of complexing salt anions seem to be a very promising group of compounds introduced into SPE. Borate esters containing oligooxyethylene groups of various length were mainly used as boron derivatives, applied as

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low molecular weight plasticizers [15–18] or incorporated into the polymer linear or cross-linked structure [19–21], for which a considerable increase in ionic conductivity was in many cases achieved. Examples of an increase in the lithium cation transference number (t_+) for SPE containing borate esters have also been described. For trialkoxyborane the highest t_+ value of 0.34 was achieved in the system with LiCF_3SO_3 [17], and for the cross-linked polymer containing boroxine rings the t_+ value lies in the 0.60–0.88 range [19]. Borane and borate derivatives containing aromatic or aliphatic substituents were also used as boron additives to polymer electrolytes [22–25]. It is assumed that the favorable effect of boron derivatives results from their Lewis acid properties, however, there are no data on the strength of these acids as well as on the strength of interaction with salt anions.

In this paper are presented the results of studies concerning the synthesis of a new group of boron derivatives comprising two fluorine atoms bonded with boron $[\text{R}_n\text{OBF}_2]_2$ and oxyethylene substituents (R_n) of various length. On the basis of NMR spectra the effect of interaction of difluoroalkoxyborane, trialkoxyborane and BF_3 with a chosen Lewis base—pyridine were compared. The ^{11}B and ^{19}F NMR spectroscopies were used to analyze the interaction of difluoroalkoxyborane and trialkoxyborane with anions of various lithium salts. The obtained fluorine derivatives of boron $[\text{R}_n\text{OBF}_2]_2$ were applied as additives to PEO-based SPE in order to evaluate their effect on the ionic conductivity and immobilization of salt anions.

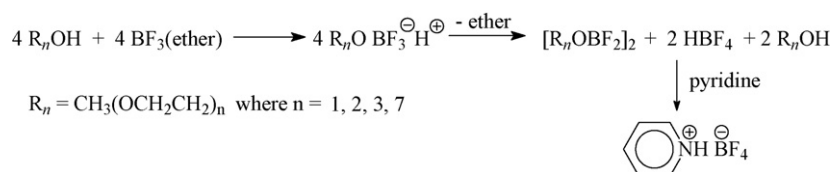
2. Experimental

2.1. Preparation of boron compounds

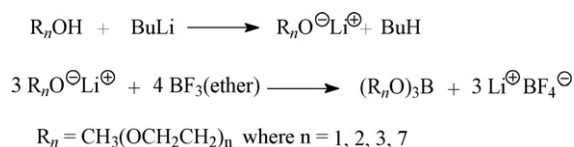
2.1.1. Preparation of difluoroalkoxyborane compounds

Difluoroalkoxyborane compounds were obtained in the reaction of oxyethylene glycols monomethyl ether of various molecular weight (R_nOH , $M_w = 76, 120, 160$ and 350 g mol^{-1} , all from Aldrich) with boron trifluoride in the form of a complex with diethyl ether (BF_3 etherate, 98% Aldrich). The most probable course of this reaction is presented in Scheme 1.

The reactions of obtaining difluoroalkoxyboranes $[\text{R}_n\text{OBF}_2]_2$ were carried out in the presence of an equimolar amount of reactants (BF_3 etherate and R_nOH) without a solvent, maintaining the reaction mixture temperature close to room temperature. All operations were carried out under the atmosphere of a neutral gas on a vacuum line. After reaction completion the diethyl ether formed in the reaction



Scheme 1. Preparation of difluoroalkoxyboranes $[\text{R}_n\text{OBF}_2]_2$ from BF_3 etherate and oxyethylene glycols monomethyl ether of various molecular weight.



Scheme 2. Preparation of trialkoxyboranes $(\text{R}_n\text{O})_3\text{B}$ in the reaction of BF_3 etherate and lithium monomethyl ether of oxyethylene alkoxides (R_nOLi) of various molecular weight.

was removed by distillation under reduced pressure. In the next step pyridine was added and the precipitated tetrafluoroboric hydride of pyridine was separated from the product and glycol mixture; the latter was then distilled off under reduced pressure. In the case of R_nOH containing $n=3$ or less ethylene oxide monomeric units, the product was purified by distillation under reduced pressure (for $n=3$, bp = $160\text{--}180^\circ\text{C}/10^{-4}$ Torr). In all the cases studied, fluoroalkoxyborane products were viscous liquids. The structure of the products was determined on the basis of ^1H , ^{11}B and ^{19}F NMR spectroscopy. An analysis of the spectra indicates a dimeric structure of the difluoroborane derivatives obtained.

2.1.2. Preparation of trialkoxyborane compounds

The trialkoxyborane derivatives were obtained in the reaction of BF_3 etherate with lithium alkoxides of analogous glycols (R_nOH) to those applied in the reaction described in Section 2.1.1. The alkoxides were obtained in the reaction of oxyethylene glycols with *n*-butyllithium (BuLi , 2.5 M solution in hexane, Aldrich) using a 1.1 molar excess of BuLi with respect to the hydroxyl groups. The reaction is described by Scheme 2.

Contrary to the reaction involving glycols, in this case trialkoxyborane $(\text{R}_n\text{O})_3\text{B}$ is formed, and LiBF_4 isolates as a side-product. On the basis of boron resonance it was estimated that the ether fraction may contain from 5 to 10% of the lithium salt in the form of a complex with trialkoxyborane.

2.1.3. Synthesis of polymer electrolytes containing difluoroalkoxyborane compounds

The electrolytes were obtained according to the standard procedure by film casting of poly(ethylene oxide) (PEO) ($M_w = 5 \times 10^6 \text{ g mol}^{-1}$), borane compound and lithium salt from a mutual solution in acetonitrile. The mixed components formed a homogeneous solution. All operations connected with the obtaining of electrolytes and performing the measurements were carried out in an atmosphere of dried argon. The solvent was removed under dynamic vacuum in two steps, first for 50 h at a vacuum of 20 Torr and then for 140 h at 10^{-3} Torr at room

temperature. In the FTIR spectrum, the signals assigned to acetonitrile completely disappeared after about the first 70 h of such drying. The solvents were dried and distilled in an argon atmosphere prior to use. The electrolytes thus obtained form flexible and homogeneous membranes, with the exception of the system with the $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ salt, for which the electrolyte was viscous.

2.2. Thermal analysis

The DSC studies were carried out with a Perkin-Elmer (Pyris-1) analyzer. The measurements were carried out in the -100 to 200°C temperature range with a heating rate of 20°min^{-1} in hermetically closed aluminum pans. An empty aluminum pan was used as a reference and the samples weights were maintained in the range of 10–15 mg.

2.3. ^1H , ^{11}B and ^{19}F NMR studies

The structures of the borane derivatives and analysis of their mixtures with lithium salts were determined by combining ^1H , ^{11}B and ^{19}F NMR spectroscopy. Spectra were recorded on a Varian Mercury 400 spectrometer in CDCl_3 , CD_3CN or dimethoxyethane (DME) with a small amount of DMSO_{d6} solution at 25°C . Boron resonance measurements were carried out in quartz tubes. The chemical shifts in ^{19}F NMR spectra were determined with respect to CF_3COOH used as an internal standard.

2.4. Electrochemical measurements

The ionic conductivity of the polymer electrolyte (SPE) membranes was determined by ac impedance spectroscopy between blocking electrodes. The sample was sandwiched between two stainless steel (SS) blocking electrodes with a diameter of 1 cm to form a symmetrical SS/SPE/SS cell and placed in a temperature-controlled oven. The impedance tests were carried out in the 1 MHz to 1 Hz frequency range using a Solartron-Schlumberger 1255 impedance analyzer.

For the determination of the lithium ion transference number (t_+) the polymer electrolyte symmetrical Li/SPE/Li hermetically sealed 2032 coin cells were assembled and tested with the use of a computer-interfaced Solartron 1260 frequency-response analyzer over the frequency range: 1 MHz to 0.1 Hz. The area of the polymer electrolyte sample was 0.96 cm^2 . The test temperature range varied from 25 to 100°C .

The lithium ion transference number (t_+) of polymer electrolytes was determined by a well-established steady-state technique introduced by Bruce et al. and Scrosati and co-workers [26,27]. The method is based on the analysis of combined dc and ac runs on the symmetrical cell with non-blocking lithium electrodes. The dc measurements of initial (I_0) and the steady-state (I_{ss}) currents were made by polarizing the test cells at voltage bias of 10 mV. The ac impedance measurements were performed before and after applying a bias, to evaluate the value of the interfacial resistance at the beginning and at the end of the test, i.e., R_0 and R_{ss} , respectively. Under these conditions, the lithium ion

transference number is given by the equation:

$$t_{\text{Li}^+} = \frac{I_{ss}(\Delta V - I_0 R_0)}{I_0(\Delta V - I_{ss} R_{ss})} \quad (1)$$

The lithium ion transference number of the membrane samples were measured between 55 and 90°C .

3. Results and discussions

3.1. NMR characterization of boron compounds

Difluoroalkoxyboranes obtained according to Scheme 1 were characterized by ^1H , ^{11}B and ^{19}F NMR spectroscopy. In the ^1H NMR spectrum of the product formed in the first reaction stage ($\text{R}_n\text{OBF}_3^-\text{H}^+$) a signal characteristic of the acidic hydrogen atoms is present at the chemical shift $\delta_{(\text{CDCl}_3)}$ 10.89 ppm. Moreover, signals at $\delta_{(\text{CDCl}_3)}$ 4.01–4.10 (m) of the hydrogen atoms in the CH_2OB group, which are strongly shifted with respect to that of R_nOH ($\delta_{(\text{CDCl}_3)}$ CH_2OH 3.50–3.53 ppm (t)) due to the negative induction effect of BF_3 , are present. The other protons in the oxyethylene groups give in the spectrum signals in the 3.6–3.8 ppm range in the form of multiplets and the OCH_3 ether group gives a signal $\delta_{(\text{CDCl}_3)}$ at 3.44–3.46 ppm. In the ^{19}F NMR spectrum this product shows a signal of chemical shift $\delta_{(\text{CDCl}_3)}$ -158.637 ppm and also one signal in the ^{11}B spectrum: $\delta_{(\text{CDCl}_3)}$ -0.638 ppm.

Under the effect of temperature or introduction to the system of a base, such as e.g. pyridine, HBF_4 isolates (with pyridine in the form of $\text{C}_5\text{H}_5\text{NH}^+\text{BF}_4^-$) and a reaction product is formed containing an alkoxide substituent and two fluorine atoms bonded with the boron atom. In the reaction, simultaneously a molecule of the initial glycol isolates. In the ^1H NMR spectrum of the product $[\text{R}_n\text{OBF}_2]_2$ no signals of acidic protons are present; the CH_2OB group in the alkoxide substituent shows a picture in the form of a triplet of chemical shift $\delta_{(\text{CDCl}_3)}$ 3.80–3.83 ppm (t). The chemical shifts of signals of the product in both boron and fluorine resonances depend considerably on the type of solvent. For solutions in CDCl_3 the fluorine signal appears at -158.6 ppm, and that of boron at ca. 0.8 ppm (Fig. 2), both in the form of singlets. In acetonitrile the fluorine signal splits showing a quartet of chemical shift in the $\delta_{(\text{CD}_3\text{CN})}$ -150.45 to -150.54 ppm range, and the boron signal remains a singlet at $\delta_{(\text{CD}_3\text{CN})}$ 5.8 ppm. In ^{19}F spectra of solutions recorded in DME (with a several percent share of DMSO_{d6}) a group of signals is observed in the range characteristic for fluorine atoms bonded with the boron atom, which may result from the cleavage of the dimeric form of the compound and its interaction with the solvent. In the ^{11}B NMR spectrum recorded in this solvent one signal is present at $\delta_{(\text{DME})}$ 3.73 ppm (Fig. 3).

In the ^1H NMR spectrum of an exemplary trialkoxyborane containing $n = 2$ oxyethylene monomeric units, the signals characteristic of the CH_2OB group show a chemical shift of $\delta_{(\text{CDCl}_3)}$ 3.69–3.72 ppm (t), and the remaining CH_2 groups show a picture of three triplets in the $\delta_{(\text{CDCl}_3)}$ 3.28–3.41 ppm range, and the OCH_3 group shows a signal at $\delta_{(\text{CDCl}_3)}$ 3.1 ppm. In the ^{11}B NMR spectrum, the trialkoxyboranes signal can be found at

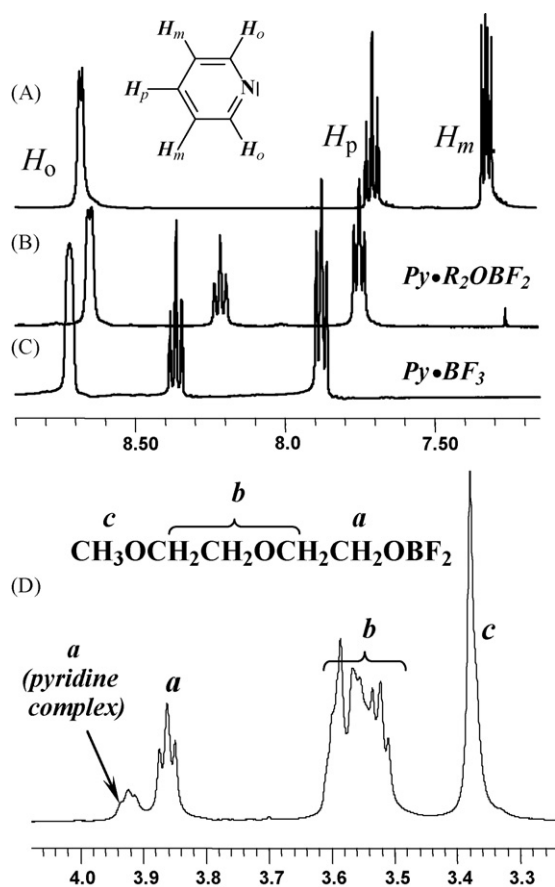


Fig. 1. ^1H NMR spectra of pyridine (A) and mixture of pyridine with: $[\text{R}_2\text{OBF}_2]_2$ containing 27% of pyridine (B, D) and BF_3 (1:1) (C) in CD_3CN .

$\delta(\text{CDCl}_3)$ 17.9 ppm in chloroform and at $\delta(\text{CD}_3\text{CN})$ about 23 ppm in acetonitrile.

3.2. NMR studies of Lewis acidic properties of trialkoxy and difluoroalkoxyborane compounds

The acidic properties of the obtained two types of derivatives, trialkoxy- and difluoroalkoxyboranes have been compared by complexation reactions with pyridine and various lithium salts. Salts comprising anions of various hardness, such as LiF , LiCl , LiI , $\text{CF}_3\text{SO}_3\text{Li}$, LiClO_4 and $(\text{CF}_3\text{SO}_2)_2\text{NLi}$ were used for the analysis. The strength of interaction of the studied boranes with the mentioned compounds of Lewis base properties was observed by ^1H , ^{11}B and ^{19}F NMR spectroscopy. In the case of lithium salts the measurements were carried out for two types of solutions containing as solvent DME and CDCl_3 and for the reaction with pyridine– CD_3CN .

The ^1H NMR spectra of solutions of $(\text{R}_n\text{O})_3\text{B}$ and pyridine mixtures in CD_3CN at an equimolar ratio as well as a twofold excess of pyridine do not show their interaction. The signals characteristic for pyridine ($\delta(\text{CD}_3\text{CN})$ 7.29–7.33 ppm (t) CH_{meta} , 7.67–7.72 ppm (t) CH_{para} , and 8.67, 8.68 ppm (d) CH_{ortho}) (Fig. 1A) did not undergo a shift, and also there are no changes in the range of the CH_2OB group signals indicating an interaction of $(\text{R}_2\text{O})_3\text{B}$ with pyridine. However, in ^1H

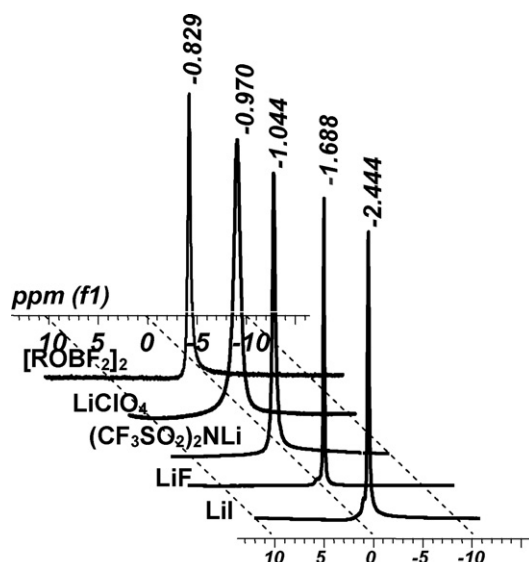


Fig. 2. Influence of various lithium salts addition on the chemical shift of the boron signal in $[\text{CH}_3(\text{OCH}_2\text{CH}_2)_2\text{OBF}_2]_2$ in ^{11}B NMR spectra of equimolar ratio solutions recorded in CDCl_3 .

NMR spectra of $[\text{R}_2\text{OBF}_2]_2$ with pyridine the changes are considerable. At a small share of pyridine, signals originating from both the complexed and free forms of $[\text{R}_2\text{OBF}_2]_2$ can be observed. The spectrum of the $[\text{CH}_3(\text{OCH}_2\text{CH}_2)_2\text{OBF}_2]_2$ (R_2OBF_2) mixture containing 27% of pyridine is presented in Fig. 1B and D.

As can be noticed, the position of pyridine signals in the complex changes and these signals occur at $\delta(\text{CD}_3\text{CN})$ 7.73–7.77 ppm (t) CH_{meta} , 8.19–8.24 ppm (t) CH_{para} , and 8.65, 8.66 ppm (d) CH_{ortho} . In the region characteristic for the alkoxy group, the signal at 3.92 can be assigned to the CH_2OB group of the difluoroborane derivative complexed form (Fig. 1D). At an equimolar and excess content of pyridine the spectrum complicates and new signals appear, difficult for interpretation, confirming, however, the strong interaction of the fluoroborane derivative with pyridine. For comparison, in Fig. 1C is presented the ^1H NMR spectrum of the BF_3 and pyridine complex. In this spectrum the pyridine signals shift more than in the case of the analyzed difluoroalkoxyborane. These signals occur at 7.86–7.89 ppm (t) CH_{meta} , 8.34–8.39 ppm (t) CH_{para} , and 8.72 ppm (d) CH_{ortho} . Comparing the acidic properties of boron derivatives by the interaction with a Lewis base—pyridine it can be found on the basis of NMR spectra that trialkoxyborane bears very weak acidic properties not observed in NMR, and the analyzed difluoroalkoxyborane shows features of a Lewis acid of weaker strength than that of BF_3 .

An analysis of the interaction of anions of various lithium salts with the borane compounds obtained was carried out on the basis of ^{11}B and ^{19}F NMR spectra for solutions of mixtures of an equimolar composition. Fig. 2 presents the spectra of solutions of $[\text{R}_2\text{OBF}_2]_2$ mixtures with lithium salts recorded in chloroform. It should be noticed that a majority of the lithium salts studied show a limited solubility in chloroform, and as a result of interaction with the analyzed difluoroalkoxyboranes they achieve considerable solubility in this solvent.

As seen in Fig. 2, in the spectrum of the $[R_2OBF_2]_2$ solution in $CDCl_3$ a singular signal of boron at -0.829 is present. In the presence of lithium salts this signal undergoes a shift towards the lower field region and this effect depends on the type of the lithium salt anion. As can be noticed $[R_2OBF_2]_2$ interacts the strongest with the iodide anion; in the presence of LiI the boron signal shifts by ca. 1.615 ppm to -2.444 ppm. The effect caused by the addition of LiF is much weaker, the chemical shift of boron in the presence of this salt is -1.688 ppm, i.e. the signal shifts by 0.859 ppm. This result is surprising, since it could be expected that the hard fluoride anion should interact stronger with boron. However, the weaker interaction may result in this case from the low polarity of chloroform, in which LiF undergoes dissociation with difficulty, and only in the dissociated form it can compete with the ether basic centers present in the boron compound alkoxy group. It is similar in the case of $LiClO_4$ and $LiN(CF_3SO_2)_2$; as can be noticed in the spectra, the hard perchlorate anion shows a weaker effect on the boron signal position than the soft imide salt anion.

The effect of the solvent on the position of signals in the boron resonance of $[R_2OBF_2]_2$ in the presence of lithium salts was compared using solutions in dimethoxyethane (DME) with 10% $DMSO_{d6}$. The ^{11}B NMR spectra in this solvent, presented in Fig. 3, confirm the interaction of fluoroalkoxyborane with lithium salts. The effect of complexation is shown by the appearance in the spectra of a new signal, also like in $CDCl_3$ shifted towards the lower field region. Moreover, the signal assigned to the non-complexed form undergoes a shift in the same direction. The boron signal in the $[R_2OBF_2]_2$ spectrum without the addition of salt, presented in Fig. 3, is not split, but as can be seen it is broadened and probably is composed of two signals. This effect is connected with the interaction of the boron compound with the solvent which has donor properties. With an increase in the ability of the added salt anions to interaction with fluoroalkoxyborane, the intensity of the non-complexed boron signal increases in the spectra. As seen in Fig. 3, for LiF nearly exclusively the complexed form is present, the chloride anions slightly weaker complex boron, whereas in the case of triflate as well as imide salt in solution the neat form prevails.

In the ^{11}B NMR spectra of trialkoxyboranes in $CDCl_3$ and DME no effect of the added lithium salt on the chemical shift was noticed. For solutions in chloroform the boron signal occurs at ca. 18 ppm, and in DME only one signal is observed at ca. 23 ppm. Therefore, the acidic properties of trialkoxyborane derivatives cannot be concluded on the basis of boron resonance.

3.3. Characterization of electrolytes comprising difluoroalkoxyborane compounds

3.3.1. Thermal properties

The obtained trialkoxy and difluoroalkoxyborane compounds were applied as additives to solid polymeric electrolytes comprising PEO as polymer matrix and 10 mol% of lithium salt. In all the cases the electrolytes obtained had the form of flexible membranes with the exception of $LiN(CF_3SO_2)_2$, for which the electrolyte was viscous. In Table 1 are presented the results of DSC measurements of electrolytes obtained involv-

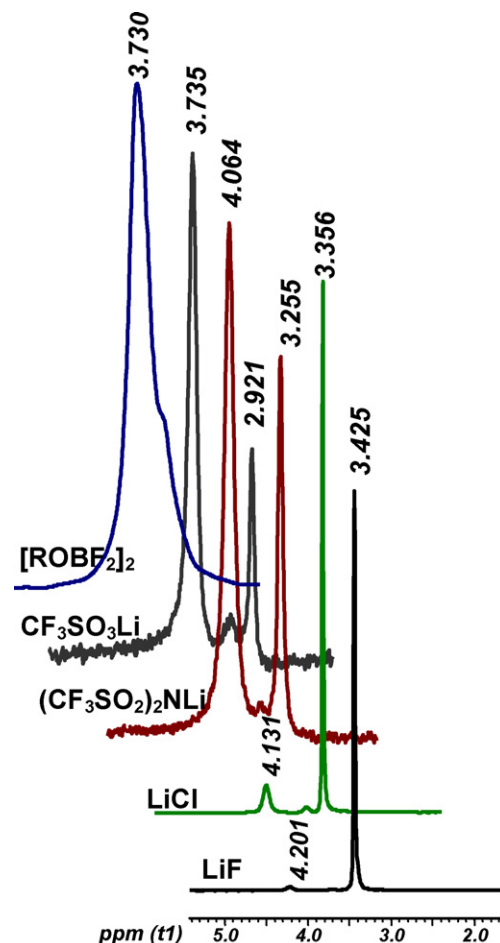


Fig. 3. ^{11}B NMR spectra of complexed and free form of $[CH_3(OCH_2CH_2)_2OBF_2]_2$ in the mixture with various lithium salts solutions at equimolar ratio, recorded in DME (+10% of $DMSO_{d6}$).

ing $[R_2OBF_2]_2$. As can be noticed, the addition of the boron compound does not limit the ability of PEO chains towards crystallization. The polymer crystalline phase content in the electrolytes increases in time, which is indicated by the repeated after 30 days result of measurement carried out for the same sample containing $LiCF_3SO_3$. The measurements performed for samples conditioned for the same period of time indicate that the greatest tendency towards crystallization is characterized by the

Table 1

DSC data of polymer electrolytes based on PEO comprising equimolar ratio of $[CH_3(OCH_2CH_2)_2OBF_2]_2$ and lithium salts^a

Salt	T_m^b (°C)	% of crystalline PEO ^c	T_g^d (°C)
LiI	56.4	38.2	-49.0
$LiN(CF_3SO_2)_2$	—	—	-47.7
$LiCF_3SO_3$	59.4	22.9	-43.4
$LiCF_3SO_3^e$	60.4	40.7	-50.2
$LiBF_4$	57.7	45.7	-38.9
LiF	65.0	79.4	-31.9

^a $Li:EO = 10$.

^b Determined from 1st heating cycle.

^c Calculated with respect to PEO content in electrolyte.

^d Determined from 1st heating cycle.

^e Measurement repeated after 30 days.

system involving LiF, for which the crystalline phase content is nearly 80%. It also has the highest glass transition temperature $T_g = -31.9^\circ\text{C}$. The electrolyte comprising $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ is completely amorphous and is characterized by low glass transition temperature of -47.7°C . At the same time it is characterized by the worse mechanical properties, it is viscous and flows under pressure. The lowest T_g of -49.0°C was found for the system with LiI, at about 40% of crystalline phase this electrolyte bears good mechanical properties, it is flexible and maintains dimensional stability under moderate pressure.

3.3.2. Electrochemical properties of electrolytes comprising difluoroalkoxyboranes

The difluoroalkoxyboranes obtained were used as additives for polymer electrolytes containing PEO as polymer matrix, and various lithium salts at an equimolar ratio of the boron compound to salt. The ionic conductivity of these systems was determined by impedance spectroscopy as a function of temperature in a system of blocking steel electrodes. Fig. 4 shows the conductivity of electrolytes involving difluoroalkoxyboranes $[\text{R}_n\text{OBF}_2]_2$ containing one or two oxyethylene monomeric units ($n = 1, 2$).

As seen in Fig. 4, the lowest ionic conductivity is shown by the electrolyte containing LiF, for which the σ course as a function of temperature is characteristic of highly crystalline systems. As appears from DSC studies, this system bears nearly 80% of PEO crystalline phase (Table 1). Its ambient temperature conductivity is $4 \times 10^{-8} \text{ S cm}^{-1}$, and above T_m the conductivity clearly increases to values of the order of $10^{-5} \text{ S cm}^{-1}$. A nearly an order higher conductivity is shown by the system containing LiBF_4 . Considerably higher values were obtained for a fully amorphous system with $\text{LiN}(\text{CF}_3\text{SO}_2)_2$, in the 10^{-5} to $10^{-4} \text{ S cm}^{-1}$ range, practically the same for $[\text{R}_n\text{OBF}_2]_2$ containing 1 or 2 oxyethylene monomeric units. Surprisingly high conductivity values, comparable with electrolytes containing an

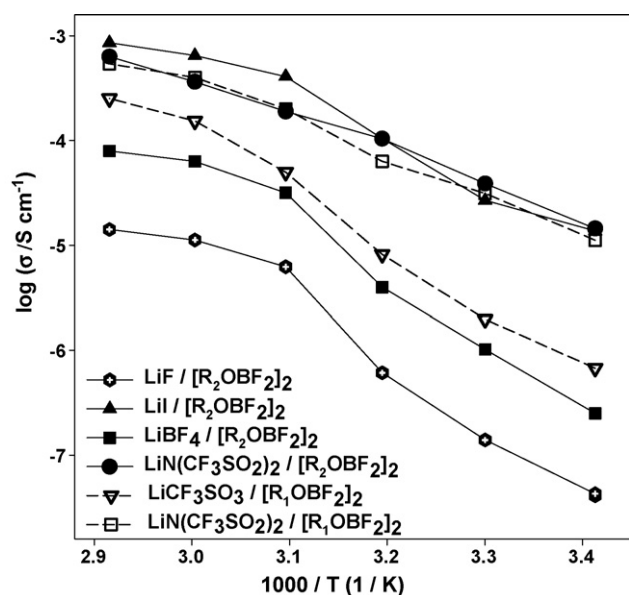


Fig. 4. Effect of the type of lithium salt on the conductivity of electrolytes based on PEO and difluoroalkoxyboranes $[\text{R}_n\text{OBF}_2]_2$ containing one or two oxyethylene monomeric units ($n = 1, 2$). $[\text{R}_n\text{OBF}_2]_2$:salt molar ratio = 1.

imide salt were achieved for systems with LiI. The Arrhenius plot for LiCF_3SO_3 lies in the range between the plots for LiF and the best conducting LiI or imide salt.

For PEO electrolytes containing $[\text{R}_2\text{OBF}_2]_2$ and two salts: LiF and LiCF_3SO_3 , electrochemical measurements were carried out in a system of lithium electrodes in three cells. The cells were preheated at 55°C for 2 h, then cooled to RT and the data were

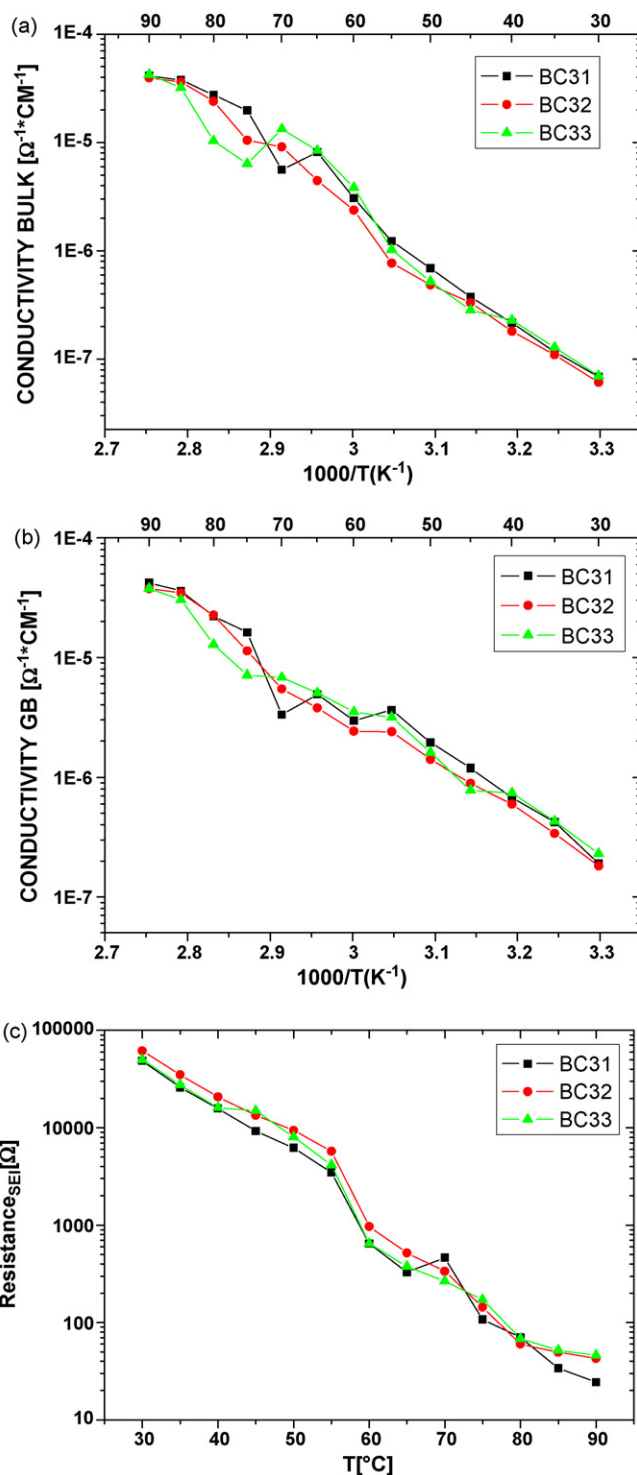


Fig. 5. Bulk (a), grain boundary (b) conductivity and SEI resistance (c) of PEO electrolytes containing $[\text{R}_2\text{OBF}_2]_2$ and LiF at 1:1 molar ratio.

collected on heating every 5 °C up to 110 °C in the frequency range from 1 MHz to 1 mHz. Fig. 5 shows the bulk and grain boundary conductivities and SEI resistance of PEO electrolytes containing $[R_2OBF_2]_2$ and LiF at a 1:1 molar ratio.

As can be seen, the bulk and grain boundary conductivities of the electrolyte containing LiF are between 10^{-8} and

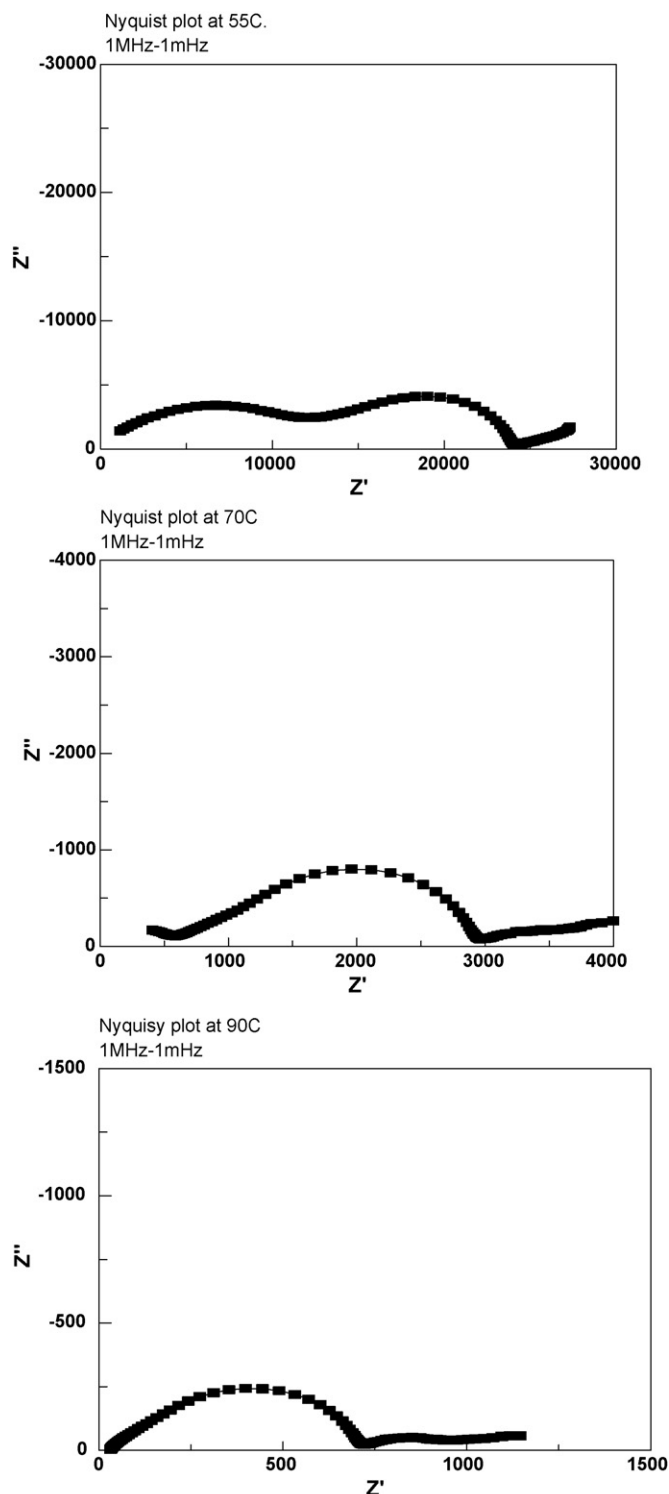


Fig. 6. Nyquist plots at low AC frequency At three temperatures (55, 70 and 90 °C) of PEO electrolytes containing $[R_2OBF_2]_2$ and LiF at 1:1 molar ratio.

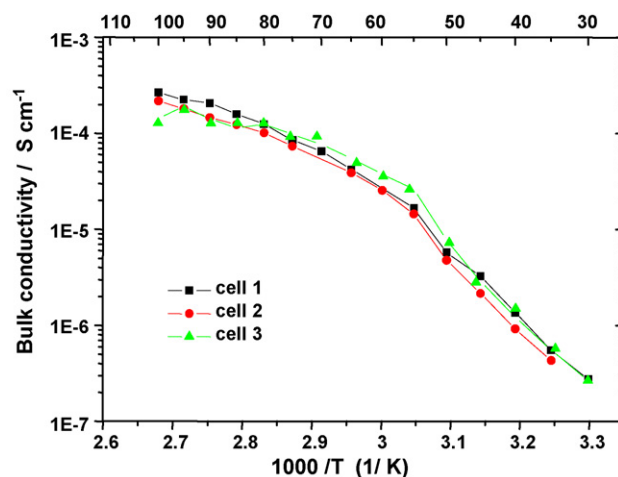


Fig. 7. Bulk conductivity of PEO electrolytes containing $[R_2OBF_2]_2$ and $LiCF_3SO_3$ at 1:1 molar ratio.

$10^{-5} \text{ S cm}^{-1}$ over the temperature range studied. Moreover, high R_{SEI} in the temperature range below 70 °C is observed; with rising temperature the resistance decreases and stabilizes at the level of 40–50 Ω. These changes are illustrated by exemplary Nyquist plots at low ac frequency presented in Fig. 6 for three temperatures: 55, 70 and 90 °C.

Analogous measurements were carried out for the electrolyte containing $LiCF_3SO_3$. Fig. 7 presents the Arrhenius plots of the Li-PEO/ $[R_2OBF_2]_2$ / CF_3SO_3Li-Li cells bulk conductivity data. Fig. 8 presents a comparison of grain boundary and bulk conductivity. As can be noticed, σ_{GB} conductivity is higher than the bulk one over the whole temperature range. The difference between σ_{bulk} and σ_{GB} increases at a temperature above melting of the PEO crystalline phase.

The lithium ion transference number values of the polymer electrolyte containing $[R_2OBF_2]_2$ and LiF or $LiCF_3SO_3$ determined by ac-dc polarization experiments are presented in Fig. 9.

As can be noticed in the diagram, for the system with LiF with a rise of temperature a rapid increase in t_+ occurs, from the value 0.58 at 55 °C to 0.82 at 90 °C. The increase in the share of lithium cations in the electric charge transport with

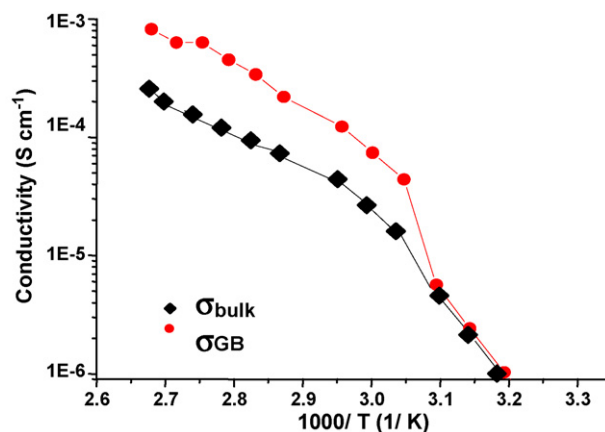


Fig. 8. Grain boundary and bulk conductivity of PEO electrolytes containing $[R_2OBF_2]_2$ and $LiCF_3SO_3$ at 1:1 molar ratio 1: 1.

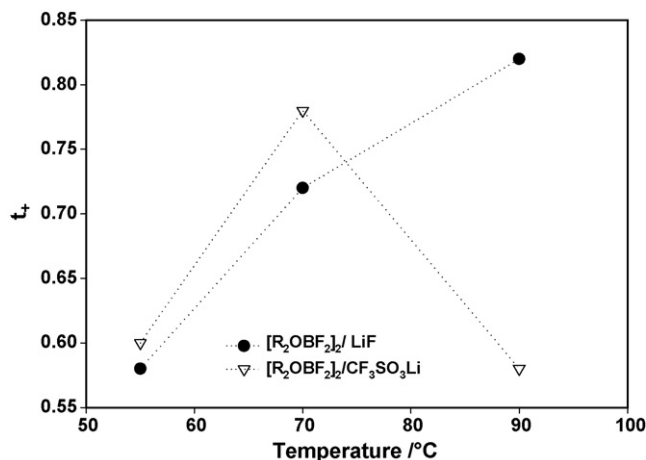


Fig. 9. Lithium transference numbers vs. temperature of polymer electrolytes comprising $[R_2OBF_2]_2$ and lithium salt LiF (●) and $LiCF_3SO_3$ (▽) at molar ratio 1:1.

a rise of temperature indicates a more efficient interaction of fluoride anions with the difluoroalkoxyborane, especially above T_m of the PEO crystalline phase ($T_m = 65^\circ C$, Table 1). Such high t_+ values show a considerable immobilization of the salt anions, which at relatively high ionic conductivity, σ at $90^\circ C$ is $4 \times 10^{-5} S cm^{-1}$, indicates high mobility of lithium cations in the polymer matrix.

For the system with lithium triflate the lithium transference number varies from 0.6 to 0.78 with a maximum at $70^\circ C$ (Fig. 9). It should be mentioned that t_+ on heating and subsequent cooling from $110^\circ C$ to room temperature were found to be very similar. In the Nyquist plots (not presented in the article) the diffusion branch is almost absent at $70^\circ C$, while it can be clearly distinguished in the plots at 55 and $90^\circ C$. The impedance behavior points to the fact of a much higher t_+ value at $70^\circ C$, than that at other temperatures. This phenomenon is difficult to explain at the present stage of studies. There is probably a complex superimposition of two effects connected with the phase changes in the electrolyte and equilibrium interaction of the salt anion with the applied Lewis acid. In general, data presented show a considerable decrease in the anion mobility when using difluoroalkoxyborane as the additive, resulting from the Lewis base–acid type interaction.

4. Conclusions

In the reaction of BF_3 etherate with oligooxyethylene glycols we obtained a new class of boron compounds—difluoroalkoxyboranes $[R_nOBF_2]_2$, which are characterized by Lewis acid properties. These properties are weaker than that of BF_3 , but clearly distinguishable in the reaction with the Lewis base used—pyridine, which was not observed for trialkoxyboranes on the basis of NMR analysis. The $[R_nOBF_2]_2$ obtained interact with lithium salts anions, and the strength of this interaction can be compared for different anions in the ^{11}B NMR spectra. It seems that small “hard” fluoride anions should complex boron easier, but probably due to the low degree of dissociation of this salt, they give smaller shifts in ^{11}B NMR

than that of iodide ions. The application of $[R_2OBF_2]_2$ as additives for PEO-based solid polymer electrolytes enables to obtain polymer electrolytes characterized by a predominant share of lithium cations in the electric charge conduction; the lithium transference number values for these systems are within the 0.6–0.8 range. The DSC data indicate that the introduction of $[R_2OBF_2]_2$ to the system in an equimolar amount with respect to the salt does not limit the tendency to crystallization of the PEO chains. The ionic conductivity of the electrolytes obtained is, therefore, an outcome of effects connected with the strength of interaction of $[R_2OBF_2]_2$ with the salt, resulting in limiting the anion mobility, degree of salt dissociation and content of the PEO crystalline phase. In our opinion the preliminary results of studies obtained are encouraging for further work in order to optimize the conducting, charge transport, and mechanical properties of electrolytes comprising fluoroalkoxyboranes.

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